Decision-time information in rolling airline recovery

Illustrative decision-time snapshot: similar delay, different operational exposure

Part 1. AOCC decision-time state

Two aircraft-linked chains assessed during turnaround

Predecessor in-block — factual

Successor off-block — unresolved

Successor take-off — unresolved

Delay priority (P^D) similar predicted take-off delay

Chain A TURN

- Turnaround opportunity
- Aircraft continuity
- Downstream exposure

Chain B TURN

- Passenger exposure
- Connection exposure
- Operating-resource burden

Consequence-based priority (P^C)

Consequence domains: Flight, Passenger, Resource

Observed: predecessor status | Infer: successor exposure | Act: shortlist → target

Part 2. Stage I screening

Delay versus consequence-based priority

Which chains enter the limited review shortlist?

Delay priority (P^D): A, B — similar delay-based priorities

Consequence (P^C): A, B — context may change selection

Same candidate set and capacity; different priority signals

Part 3. Stage II local recovery

Within one shortlisted chain: effective local-recovery target

Determine whether recovery is worthwhile and choose u

Feasible target set — recovery headroom

(u=0): no recovery

(u>0): effective local-recovery target

Information value depends on the supported decision